Sajid Mostafiz Noor

Mohammadpur, Dhaka — +880-1815-458659 — sajidmostafiznoor@gmail.com — GitHub — Portfolio

Education

Bangladesh University of Engineering and Technology (BUET) 2022 – 2026
B.Sc. in Computer Science and Engineering (CSE)
CGPA: 3.70 / 4.00

Projects

BUET Fantasy Premier League (Live) July 2025 – September 2025

- Built a fantasy sports platform modeled after the Premier League FPL system for managing fantasy teams based on real players.
- Developed RESTful APIs, scoring algorithms, and real-time leaderboards.
- Enabled users to manage teams, make team changes, make transfers, view player statistics, fixtures, and match results.
- Deployed the platform on Vercel with optimized performance for better user experience.

ExamCraft – AI-driven Exam Preparation Platform (GitHub) February 2025 – August 2025

- Developed an AI-powered platform for quizzes, flashcards, notes, and timed mock exams for topics/subjects.
- Enabled AI-based question and flashcard generation from study materials.
- Added spaced-repetition learning and performance analytics for custom recommendations and progress tracking.
- Automated build and deployment processes with GitHub Actions and deployed the containerized application on a virtual machine.

C++ Compiler (GitHub) January 2024 - March 2024

- Built a compiler from scratch in C++ using Flex (lexical analysis) and Bison (parsing).
- Designed grammar rules, symbol table management, and semantic analysis with error detection and reporting.
- Generated intermediate three-address code as a machine-independent representation for execution and optimization.

Grades Information System (GitHub) June 2023 – September 2023

- Designed a role-based academic management system with modular student, teacher, and admin services ensuring controlled data access.
- Engineered core backend logic for course enrollment with prerequisite validation, grade management, and result processing.
- Created an admin dashboard with CRUD operations for managing students, teachers, courses, notices, and exam schedules, and implemented database logging to ensure data consistency and track system changes.

Movie Database Manager (GitHub) August 2022 - October 2022

- Built a JavaFX-based system to manage and browse movie records with different filters (title, year, genre, runtime, revenue, recent releases, etc.).
- Implemented search, sorting, top/recent views, and adding new movies through dedicated UI workflows.
- Integrated a Java socket-based client-server architecture for centralized data handling, incorporated file I/O to store and reload movie records across sessions, and structured the app with modular OOP classes (Movie, MovieList, controllers).

Research Experience & Publications

Real Time Machine Learning Based Detection of Aggressive Driving using Feature Optimization 2025 – 2026
*Ibnul Islam, **MD Sajid Mostafiz Noor**, Ashikur Rahman, Raqeebir Rab, Abderrahmane Leshob*
Accepted in International Conference on Intelligent Multimedia, Networking, and Security, Atlanta, Georgia, USA

- Developed a real-time aggressive driving detection system using optimized sensor features.
- Used three feature selection algorithms (Feature Inclusion, Feature Exclusion, and Feature Inclusion with Group Sequencing Algorithms) to reduce input dimensionality.
- Achieved 98.54% accuracy with 66% reduction in the number of features, along with a feature-sparsity aware inference mechanism for reliable prediction under missing sensor data to improve efficiency and deployability.

Forecasting Occupational Survivability of Rickshaw Pullers Under a Changing Climate 2023 – 2025

Rahaman, M., Hasana, M., Rahman, S. S., ***Noor, M. S. M.****, Abedin, R. R., Tahmid, M. T., Parris, D. W., Choudhury, T., Al Islam, A. B. M., & Rahman, T*

ACM IMWUT — doi: 10.1145/3770712

- Analyzed physiological stress on rickshaw-pullers under extreme heat using wearable sensor data.
- Applied Bayesian regression to model occupational heat-risk and survivability and integrated CMIP6 climate projections to forecast long-term climate impact.
- Conducted data preprocessing, statistical modeling, result validation; performed thematic analysis using qualitative data collected from structured interviews with rickshaw pullers.

Next-Generation Computing Lab, BUET CSE 2023 – Present

- Collected data using wearable sensors, performed thematic analysis, and surveys for ongoing lab projects.
- Analyzed and visualized data on the heat impact on rickshaw pullers including data preprocessing and interpretation of results.

Skills

Programming: Python, Java, C, C++, JavaScript, TypeScript
Frontend: React, Next.js, Tailwind CSS, JavaFX
Backend & APIs: Node.js, Express.js, NestJS, FastAPI, REST APIs
Databases & DevOps: MySQL, MongoDB, Docker, Git, GitHub Actions, Vercel
AI & ML: Machine Learning, Deep Learning, Scikit-learn, PyTorch, TensorFlow